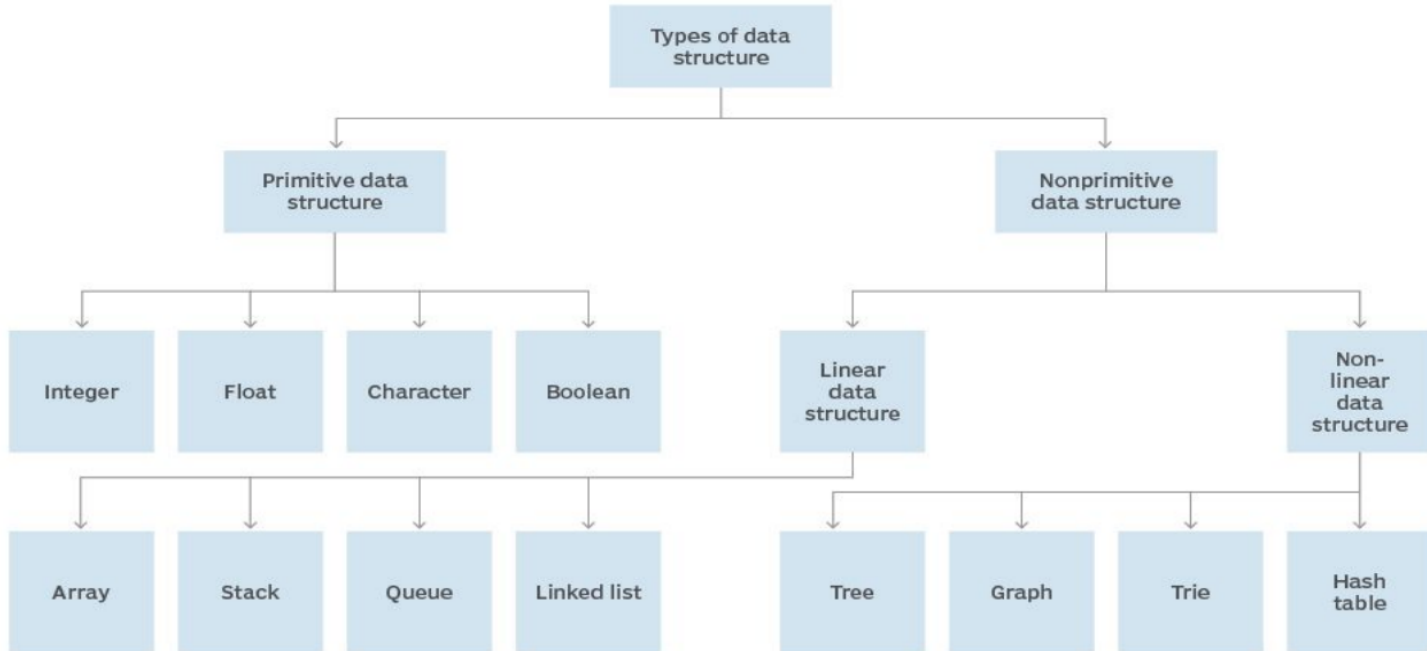




Data Strcture

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Data structure hierarchy





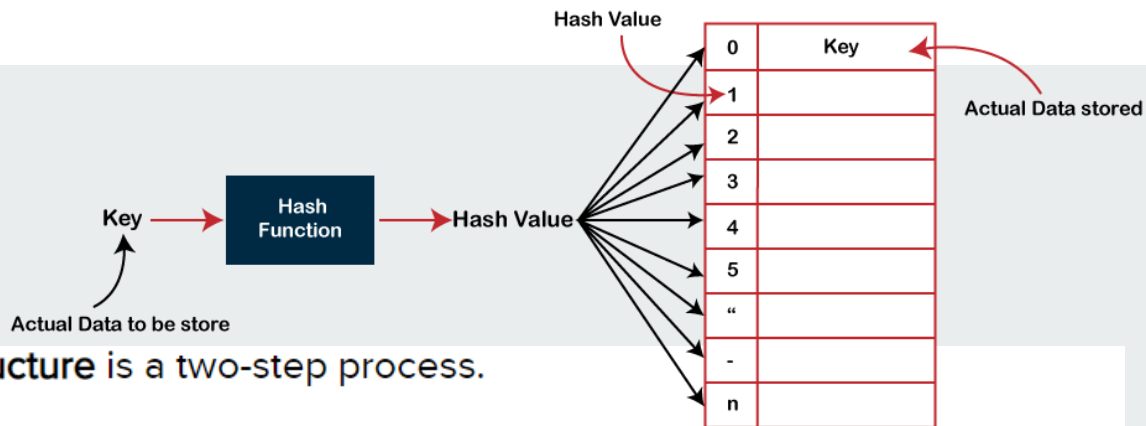
Hashing

What is hashing

Hashing in the **data structure** is a technique of mapping a large chunk of data into small tables using a hashing function. It is also known as the message digest function. It is a technique that uniquely identifies a specific item from a collection of similar items.

It uses hash tables to store the data in an array format. Each value in the array has assigned a unique index number. Hash tables use a technique to generate these unique index numbers for each value stored in an array format. This technique is called the hash technique.

You only need to find the index of the desired item, rather than finding the data. With indexing, you can quickly scan the entire list and retrieve the item you wish. Indexing also helps in inserting operations when you need to insert data at a specific location. No matter how big or small the table is, you can update and retrieve data within seconds.



Hashing in a data structure is a two-step process.

1. The hash function converts the item into a small integer or hash value. This integer is used as an index to store the original data.
2. It stores the data in a hash table. You can use a hash key to locate data quickly.

Hash Table

Hashing in data structure uses hash tables to store the key-value pairs. The hash table then uses the hash function to generate an index. Hashing uses this unique index to perform insert, update, and search operations.



Hash Function

The hash function in a data structure maps arbitrary size of data to fixed-sized data. It returns the following values: a small integer value (also known as hash value), hash codes, and hash sums.

hash = hashfunc(key)

index = hash % array_size

The has function must satisfy the following requirements:

- A good hash function is easy to compute.
- A good hash function never gets stuck in clustering and distributes keys evenly across the hash table.
- A good hash function avoids collision when two elements or items get assigned to the same hash value.

Let us understand **hashing in a data structure** with an example. Imagine you need to store some items (arranged in a key-value pair) inside a hash table with 30 cells.

The values are: (3,21) (1,72) (40,36) (5,30) (11,44) (15,33) (18,12) (16,80) (38,99)

The hash table will look like the following:

Serial Number	Key	Hash	Array Index
1	3	$3\%30 = 3$	3
2	1	$1\%30 = 1$	1
3	40	$40\%30 = 10$	10
4	5	$5\%30 = 5$	5
5	11	$11\%30 = 11$	11
6	15	$15\%30 = 15$	15
7	18	$18\%30 = 18$	18
8	16	$16\%30 = 16$	16



Examples of Hashing in Data Structure

The following are real-life examples of **hashing in the data structure** –

- In schools, the teacher assigns a unique roll number to each student. Later, the teacher uses that roll number to retrieve information about that student.
- A library has an infinite number of books. The librarian assigns a unique number to each book. This unique number helps in identifying the position of the books on the bookshelf.



Home Task

Write two programs on hashing concept